India Job Market 2024–2026 – Data Cleaning Project

1. Project Title

India Job Market 2024–2026 – Data Cleaning and Preprocessing

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2. Project Objective

The objective of this project is to clean, preprocess, and transform the India Job Market 2024–2026 dataset into a consistent and analysis-ready format.

The project focuses on identifying and handling common data-quality problems such as:

- **Missing values**
- **Duplicate records**
- **Incorrect data types**
- **Inconsistent text formatting**
- **Potential outliers**
- **Categorical variables0zation, Business Intelligence, and Machine Learning projects.**

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3. Dataset Description

The dataset contains information about job opportunities in India during the period 2024–2026.

Dataset Size

Description	Value
Total Records	5,000
Original Columns	17
Cleaned Records	5,000
Missing Values	0
Duplicate Records	0

Main Dataset Columns

Column	Description	Data Type
Job_ID	Unique job identifier	Identifier
Job_Title	Job position/title	Categorical
Company	Hiring company	Categorical

Company_Type	**Type of company**	**Categorical**
Industry	**Industry sector**	**Categorical**
City	**Job location/city**	**Categorical**
Location_Tier	**Classification of city/location**	**Categorical**
Experience_Level	**Required experience level**	**Categorical**
Job_Type	**Type of employment**	**Categorical**
Work_Mode	**Remote, hybrid, or on-site**	**Categorical**
Salary_LPA	**Salary offered in Lakhs Per Annum**	**Numeric**
Skills_Required	**Skills required for the job**	**Categorical/Text**
Education_Required	**Required educational qualification**	**Categorical**
Openings	**Number of available positions**	**Numeric**
Applicants	**Number of applicants**	**Numeric**
Company_Rating	**Company rating**	**Numeric**
Date_Posted	**Date when job was posted**	**Date**

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4. Technologies Used

Programming Language

- **Python 3.x**

Libraries

- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**

Development Environment

- **Jupyter Notebook**
- **Google Colab**
- **Visual Studio Code (optional)**

File Formats

- **CSV**

- **Excel**
- **Jupyter Notebook**

5. Dataset

- Raw file: `india_job_market_2024_2026.csv`

- Raw rows: 5000

- Raw columns: 17

- Cleaned rows: 5000

- Cleaned columns: 21

6. Tools

Python 3.x, Pandas, NumPy, Seaborn, Matplotlib, Jupyter Notebook/Google Colab.

7. Cleaning Performed

1. Loaded and explored the CSV using `head()`, `info()`, `describe()`, shape and data-type checks.

2. Checked missing values and generated a heatmap.

3. Applied column-appropriate missing-value handling.

4. Detected and removed exact duplicate rows.

5. Standardized column names to snake_case.

6. Corrected numeric and date data types.

7. Standardized whitespace in text/categorical fields.

8. Detected potential outliers with the IQR method.

9. Flagged potential outliers instead of automatically deleting potentially valid job-market records.

10. Created a separate one-hot encoded dataset for selected categorical columns.

11. Validated missing values, duplicates, types and summary statistics.

12. Exported cleaned and encoded CSV files.

8. Missing-Value Strategy

Numeric fields use the median when missing. Categorical fields use the mode. Date fields use a median date when required.

9. Outlier Strategy

IQR was applied to Salary LPA, Openings, Applicants and Company Rating. Outliers are retained and flagged because extreme values may represent legitimate observations.

10. Project Deliverables

The project contains the following deliverables:

Task_1_India_Job_Market_Deliverables/

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├── data/

| ├── raw/

| | └── india_job_market_2024_2026.csv

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| └── cleaned/

| ├── india_job_market_2024_2026_cleaned.csv

| └── india_job_market_2024_2026_encoded.csv

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├── notebooks/

| └── Task_1_India_Job_Market_Data_Cleaning.ipynb

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├── reports/

| ├── Data_Cleaning_Report.md

| ├── before_after_comparison.csv

| ├── missing_value_report.csv

| └── outlier_iqr_report.csv

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├── visualizations/

| ├── missing_values_heatmap.png

| ├── duplicates_before_after.png

| ├── boxplot_salary_lpa.png

| ├── boxplot_openings.png

| ├── boxplot_applicants.png

| ├── boxplot_company_rating.png

| └── iqr_outlier_counts.png

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└── README.md

11. How to Run the Project

Step 1

Install Python 3.x.

Step 2

Install the required libraries:

pip install pandas numpy matplotlib seaborn openpyxl

Step 3

Open the Jupyter Notebook:

jupyter notebook

Step 4

Open:

Task_1_India_Job_Market_Data_Cleaning.ipynb

Step 5

Run all cells from top to bottom.

Step 6

Review the generated:

- **Cleaned dataset**
- **Encoded dataset**
- **Reports**
- **Visualizations**
- **Data-quality results**

12. Major Findings

The raw dataset contains 5,000 job records across 17 original fields.

The dataset had:

- **No missing values**
- **No duplicate records**

- **Appropriate information for numerical and categorical analysis**
- **Potential outliers in salary, openings, and applicant counts**

The IQR method identified potential extreme observations, but these were retained and flagged because they may represent genuine job-market conditions.

The dataset was standardized and converted into an analysis-ready format.

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13. Business Use Cases

The cleaned dataset can be used for further analysis such as:

- **Most in-demand job roles**
- **Salary comparison by industry**
- **Salary comparison by city**
- **Remote vs on-site job analysis**
- **Experience level analysis**
- **Most requested technical skills**
- **Company rating analysis**
- **Applicant-to-opening ratio**
- **Job posting trends over time**
- **Industry-wise hiring analysis**
- **Location-wise job opportunities**

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14 . Future Improvements

Future versions of the project can include:

- **Exploratory Data Analysis (EDA)**
- **Interactive dashboards**
- **Salary distribution analysis**
- **Job demand forecasting**
- **Skill-demand analysis**
- **Machine-learning models**
- **Job recommendation systems**
- **City-wise employment analysis**
- **Industry-wise salary prediction**

15. Conclusion

The India Job Market 2024–2026 dataset was successfully cleaned and prepared for further data analysis.

The project addressed the major data-quality requirements including:

- **Dataset exploration**
- **Missing-value validation**
- **Duplicate validation**
- **Column standardization**
- **Data-type correction**
- **Text cleaning**
- **IQR-based outlier detection**
- **Categorical encoding**
- **Final validation**
- **Cleaned dataset export**

The resulting dataset is suitable for Data Analysis, Data Visualization, Business Intelligence, and Machine Learning applications.

16 . Author

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Project: Data Cleaning Project – Task 1
Dataset: India Job Market 2024–2026
Tools: Python, Pandas, NumPy, Matplotlib, Seaborn
Environment: Jupyter Notebook / Google Colab